How-To Library

Informatica

Address Verification Best Practices for Denmark

© Copyright Informatica LLC 2022. Informatica and the Informatica logo are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html.

Abstract

This document tells you how to connect the elements in a Denmark address to the appropriate address elements in Informatica Address Verification. This document also describes best practices that you can implement when you verify addresses in Denmark.

Supported Versions

- Informatica Address Verification 3.0 and later
- Informatica Address Verification (On-Premises) 5.x

Table of Contents

Introduction. . . . 2
Process Modes. . . . 2
Character Set. . . . 3
Address Format and Address Structure. . . . 3
Key Address Elements. . . . 3
Mapping Input Address Data to Address Elements. . . . 3
Input and Result XML in Address Verification (On-Premises). . . . 4
Input and Result XML in Address Verification. . . . 4
Address Enrichments. . . . 6
Reference Address Databases. . . . 6
Additional Documentation Resources. . . . 7

Introduction

Informatica Address Verification can validate and enhance addresses in Denmark. This how-to article provides you with configuration information and best practices that you can follow to get the most from the addresses. The best practices include advice on how to select the appropriate XML elements for Denmark address data.

To verify the quality of an input address, Address Verification compares the address information to the data in one or more reference data files. You can configure Address Verification to analyze the different types of address information in the input data. You must map each item of address information to an appropriate element in Address Verification.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies determines the elements that you need.

Process Modes

Informatica Address Verification supports the following process modes for Denmark addresses:

- BATCH
- COUNTRY RECOGNITION
- FAST COMPLETION
- INTERACTIVE
- PARSE

Character Set

Denmark reference address databases store address data in the Latin-1 script.

Address Format and Address Structure

You improve the performance of Address Verification when you define the input address in the format that the postal service expects.

The Denmark postal service recognizes the following structure for a postal address:

```
Line 1: RECIPIENT OR CONTACT NAME
Line 2: STREET NAME HOUSE NUMBER
Line 3: POSTAL CODE CITY [SORTING CODE]
Line 4: COUNTRY
```

The following sample address illustrates the structure:

```
AMALIEGADE 39
1256 KØBENHAVN K
DNK
```

Key Address Elements

Address Verification is unlikely to verify an address successfully if one of the following items is absent from the address:

- Locality
- Postal Code
- Street

Mapping Input Address Data to Address Elements

To verify an address, map the input address information to appropriate elements in Address Verification.

The following table shows how you can map the input address to the address elements:

Information Type	Address Information	Address Elements
Street Name	AMALIEGADE	Street
House Number	39	HouseNumber
Postal Code	1256	PostalCode
City name and optional sorting code	KØBENHAVN K	Locality
Country	DNK	Country

Input and Result XML in Address Verification (On-Premises)

The following example shows how you can map the address information to the XML address elements in Address Verification (On-Premises).

You enter the following input address:

```
<InputData>
    <AddressElements>
        <Country Item="1" Type="NAME">DNK</Country>
    </AddressElements>
    <AddressLines>
        <FormattedAddressLine Line="1">AMALIEGADE 39</FormattedAddressLine>
        <FormattedAddressLine Line="2">1256 KØBENHAVN K</FormattedAddressLine>
    </AddressLines>
</InputData>
```

Informatica Address Verification (On-Premises) returns the following result:

```
<?xml version="1.0" encoding="UTF-16"?>

<!-- EngineVersion="5.14.1.47633" -->

<Result     ProcessStatus="V4"
     ModeUsed="BATCH"
     Count="1"
     CountOverflow="NO"
     CountryISO3="DNK"
     PreferredScript="DATABASE"
     PreferredLanguage="DATABASE">

<ResultData ResultNumber="1"
     MailabilityScore="5"
     ResultPercentage="100.00"
     ElementResultStatus="F0F080F0F000000000E0"
     ElementInputStatus="60600060600000000060"
     AddressResolutionCode="00000000000000000000"
     ExtElementStatus="00000000000000000000"
     ElementRelevance="10100010100000000010"
     AddressType="S"
     LanguageISO3="XXX">
    <AddressElements>
        <Country Type="NAME_EN" Item="1">Denmark</Country>
        <Locality Item="1">København K</Locality>
        <PostalCode Item="1">1256</PostalCode>
        <Province Item="1">Hovedstaden</Province>
        <Street Item="1">Amaliegade</Street>
        <Number Item="1">39</Number>
    </AddressElements>
    <AddressLines>
        <DeliveryAddressLine Line="1">Amaliegade 39</DeliveryAddressLine>
        <CountrySpecificLocalityLine Line="1">1256 København K</CountrySpecificLocalityLine>
        <FormattedAddressLine Line="1">Amaliegade 39</FormattedAddressLine>
        <FormattedAddressLine Line="2">1256 København K</FormattedAddressLine>
        <FormattedAddressLine Line="3">Denmark</FormattedAddressLine>
    </AddressLines>
    <AddressComplete>Amaliegade 39
1256 København K
Denmark</AddressComplete>
</ResultData>
</Result>
```

Input and Result XML in Address Verification

The following example shows how you can verify a Denmark address in a SOAP call to Address Verification.

You submit the following request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:v4="http://
validator5.AddressDoctor.com/Webservice5/v4">
```

```
   <soapenv:Header/>
   <soapenv:Body>
      <v4:Process>
         <v4:login>****</v4:login>
         <v4:password>****</v4:password>
         <v4:parameters>
            <v4:ProcessMode>Batch</v4:ProcessMode>
            <v4:ValidationParameters>
            </v4:ValidationParameters>
         </v4:parameters>
         <v4:addresses>
            <v4:Address>
               <v4:FormattedAddress>
                  <v4:string>AMALIEGADE 39</v4:string>
                  <v4:string>1256 KØBENHAVN K</v4:string>
                  <v4:string>DNK</v4:string>
               </v4:FormattedAddress>
            </v4:Address>
         </v4:addresses>
         <v4:enrichments>
         </v4:enrichments>
      </v4:Process>
   </soapenv:Body>
</soapenv:Envelope>
```

Informatica Address Verification returns the following response:

```
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
   <soap:Body>
      <ProcessResponse xmlns="http://validator5.AddressDoctor.com/Webservice5/v4">
         <ProcessResult>
            <StatusCode>100</StatusCode>
            <StatusMessage>OK</StatusMessage>
            <Results>
               <Result>
                  <ProcessStatus>V4</ProcessStatus>
                  <CountryISO3>DNK</CountryISO3>
                  <ResultDataSet>
                     <ResultData>
                        <ResultNumber>1</ResultNumber>
                        <MailabilityScore>5</MailabilityScore>
                        <ResultPercentage>100.00</ResultPercentage>
                        <ElementInputStatus>6060006060000000060</ElementInputStatus>
                        <ElementResultStatus>F0F080F0F000000000E0</ElementResultStatus>
                        <ElementRelevance>1010001010000000010</ElementRelevance>
                        <ExtElementStatus>00000000000000000000</ExtElementStatus>
                        <AddressResolutionCode>000000000000000000000</AddressResolutionCode>
                        <AddressType>S</AddressType>
                        <LanguageISO3>???</LanguageISO3>
                        <Address>
                           <Street>
                              <string>Amaliegade</string>
                           </Street>
                           <HouseNumber>
                              <string>39</string>
                           </HouseNumber>
                           <Locality>
                              <string>København K</string>
                           </Locality>
                           <PostalCode>
                              <string>1256</string>
                           </PostalCode>
                           <Province>
                              <string>Hovedstaden</string>
                           </Province>
                           <Country>
                              <string>DENMARK</string>
                           </Country>
                           <DeliveryAddressLines>
                              <string>Amaliegade 39</string>
```

```
                              </DeliveryAddressLines>
                              <CountrySpecificLocalityLine>
                                 <string>1256 København K</string>
                              </CountrySpecificLocalityLine>
                              <FormattedAddress>
                                 <string>Amaliegade 39</string>
                                 <string>1256 København K</string>
                                 <string>DENMARK</string>
                              </FormattedAddress>
                              <AddressComplete>Amaliegade 39;1256 København K;DENMARK</
AddressComplete>
                           </Address>
                        </ResultData>
                     </ResultDataSet>
                  </Result>
               </Results>
            </ProcessResult>
         </ProcessResponse>
      </soap:Body>
   </soap:Envelope>
```

Note: The Province item in the above address identifies Hoved staden as the province for the address. Postal delivery does not require the province information.

Address Enrichments

You can configure Informatica Address Verification to add enrichments to verified Denmark addresses.

The following table lists the types of enrichment that you can add to a Denmark address:

Enrichment Type	Description
CAMEO	You can retrieve CAMEO profiles as enrichments for valid Denmark addresses. CAMEO profiles are consumer classification systems that provide socio-economic and demographic indicators about the residents at the address.
Standard Geocoding	You can retrieve standard, interpolated geocoordinates for valid Denmark addresses.
Arrival Point Geocoding	You can retrieve arrival point geocoordinates for valid Denmark addresses.

Reference Address Databases

Informatica updates the Denmark reference address databases quarterly. You can download the latest database packages from the Informatica Database Download portal. For news about reference address databases and software updates for Address Verification, subscribe to the Informatica DaaS Newsletter from Informatica.

Informatica Address Verification provides the following reference databases for Denmark addresses:

- `DNK5BI.MD`. Batch and interactive data.
- `DNK5CA.MD`. CAMEO data.
- `DNK5FC.MD`. Fast completion data.
- `DNK5GC.MD`. Standard geocoding data.
- `DNK5GCAP.MD`. Arrival point geocoding data.

Note: You download databases for Informatica Address Verification (On-Premises). You do not need to download databases for Informatica Address Verification, as Informatica Address Verification uses a cloud interface.

Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, click the following links:

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor-cloud

Author

Shahani Natalia Mendonca

Acknowledgements

The author would like to thank Amarpal Kaur Sohi, David Handy, and Farhan Ashraf for their assistance.